BITS Pilani

Pilani|Dubai|Goa|Hyderabad

Artificial and Computational Intelligence

AIMLCZG557

ACI Team

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AIMLCZG557 – CS#11-12

Probabilistic Representation and Reasoning

Agenda for CS #11-12

1. Recap of CS#10
2. Brief on Probability (Conditional Probability)
3. Bayes Theorem
4. Bayesian Networks
5. Inference with Bayesian Networks
6. Exact Inference
   - By Enumeration
   - Variable Elimination
7. Approximate Inference
8. Exercises

Course Progress

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M1	Introduction to AI
M2	Problem Solving Agent using Search
M3	Game Playing
M4	Knowledge Representation using Logics
M5	Probabilistic Representation and Reasoning
M6	Reasoning over time
M7	Multiagent Decision Making
M8	Ethics in AI

Reasoning

- Monotonic Reasoning
- Non- Monotonic Reasoning

Monotonic	Non-Monotonic
Consistent	Relaxed Consistency
Complete Knowledge	Incomplete Knowledge
Static	Dynamic
Discrete	Continuous & Learning Agent
Predicate Logic	Probabilistic Model

Need for Reasoning with Uncertainty

- The world is full of uncertainty
  - chance nodes/sensor noise/actuator error/partial info..
  - Logic is brittle
    - can't encode exceptions to rules
    - can't encode statistical properties in a domain
  - Computers need to be able to handle uncertainty
- Probability: new foundation for AI (& CS!)
- Massive amounts of data around today
  - Statistics and CS are both about data
  - Statistics lets us summarize and understand it
  - Statistics is the basis for most learning
- Statistics lets data do our work for us

Uncertainty

You can reach Bangalore Airport from MG Road within 90 mins if you go by route A.

- There is uncertainty in this information due to partial observability and non determinism
- Agents should handle such uncertainty

Previous approaches like Logic represent all possible world states

Such approaches can't be used as multiple possible states need to be enumerated to handle the uncertainty in our information

Uncertainty

You can reach Bangalore Airport from MG Road within 90 mins if you go by route A.

Road Block	Festival Season	Weekend	Observation (20)	Prob
F	F	F	12	0.6
F	F	T	3	0.15
F	T	F	2	0.1
F	T	T	2	0.1
T	F	F	0	0
T	F	T	0	0
T	T	F	1	0.05
T	T	T	0	0
				=1

Probability Basics (Self Study)

- Begin with a set S: the sample space
  - e.g., 6 possible rolls of a die.
- x ϵ S is a sample point/possible world/atomic event
- A probability space or probability model is a sample space with an assignment P(x) for every x s.t. $0 \leq P(x) \leq 1$ and $\Sigma P(x) = 1$
- An event A is any subset of S
  - e.g. A= ‘die roll < 4’
- A random variable is a function from sample points to some range, e.g., the reals or Booleans

Types of Probability Spaces (Self Study)

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Propositional or Boolean random variables

e.g., $Cavity$ (do I have a cavity?)

Discrete random variables (*finite* or *infinite*)

e.g., $Weather$ is one of $\langle sunny, rain, cloudy, snow\rangle$

$Weather = rain$ is a proposition

Values must be exhaustive and mutually exclusive

Continuous random variables (*bounded* or *unbounded*)

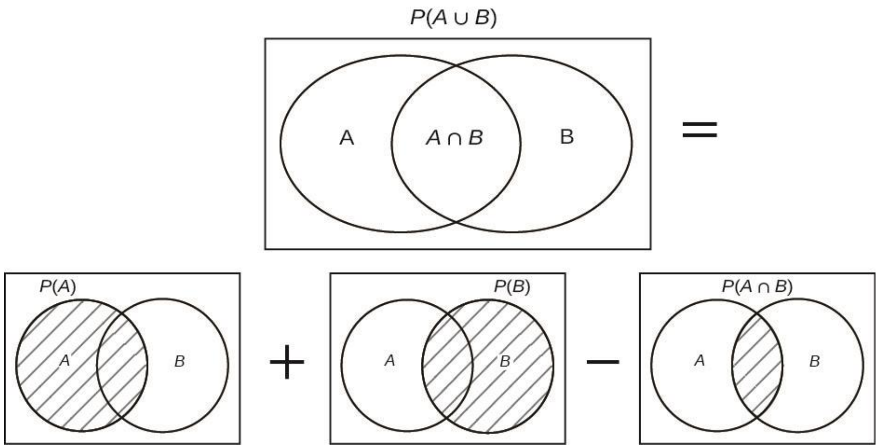
e.g., $Temp = 21.6$; also allow, e.g., $Temp < 22.0$.

Arbitrary Boolean combinations of basic propositions

Axioms of Probability Theory (Self Study)

- All probabilities between 0 and 1
  - $0 \leq P(A) \leq 1$
  - $P(true) = 1$
  - $P(false) = 0$
- The probability of disjunction is:

$$P(A \vee B) = P(A) + P(B) - P(A \wedge B)$$



Conditional Probability (Self Study)

- Conditional or posterior probabilities

e.g., P(*cavity*| *toothache*) = 0.8

i.e., given that *toothache* is all I know there is 80% chance of cavity

- Notation for conditional distributions:

P(*Cavity* | *Toothache*) = 2-element vector of 2-element vectors)

- If we know more, e.g., *cavity* is also given, then we have

P(*cavity* |*toothache, cavity*) = 1

- New evidence may be irrelevant, allowing simplification:

P(*cavity* |*toothache, sunny*) = P(*cavity* | *toothache*) = 0.8

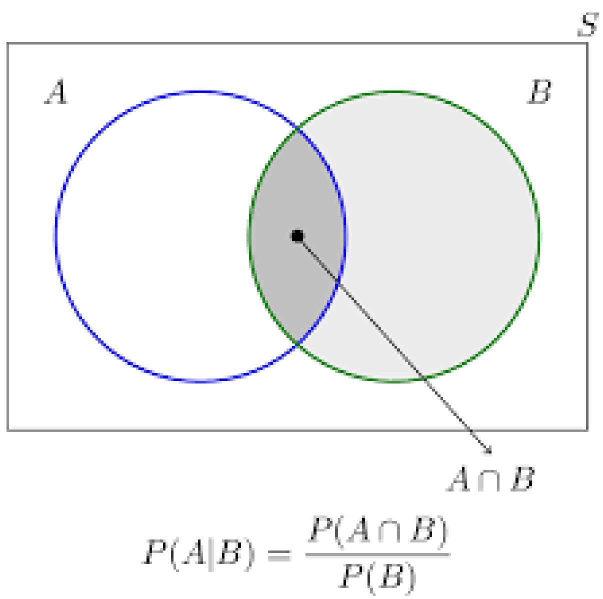
- This kind of inference, sanctioned by domain knowledge, is crucial

Conditional Probability (Self Study)

P(A | B) is the probability of A given B

Assumes that B is the only info known.

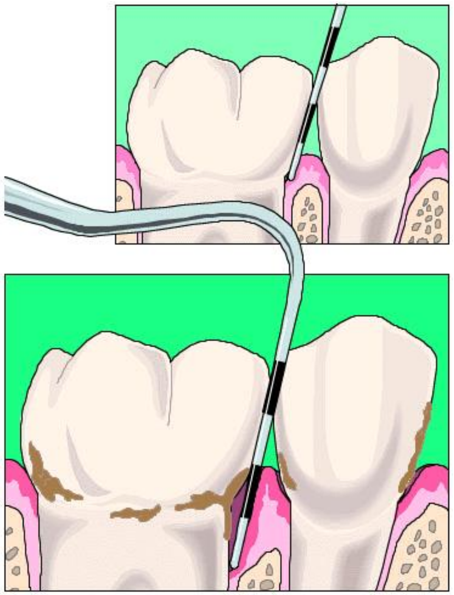
Defined by: $$P(A \mid B) = \frac{P(A \wedge B)}{P(B)}$$



Chain Rule/Product Rule (Self Study)

$$P(X_1, \ldots, X_n) = P(X_n | X_1..X_{n-1}) P(X_{n-1} | X_1..X_{n-2}) \ldots P(X_1)$$
$$= \prod P(X_i | X_1,..X_{i-1})$$

Dilemma at the Dentist's (Self Study)



What is the probability of a cavity given a toothache?
What is the probability of a cavity given the probe catches?

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

Joint Probability Distribution:

- ● Table must sum to 1
- ● Size of table is a concern

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

To reason on any proposition φ, sum the atomic events where it is true: $P(\varphi) = \Sigma_{\omega:\omega \models \varphi} P(\omega)$

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

<u>Compute P(toothache):</u>

P (toothache)

= 0.108 + 0.012 + 0.016 + 0.064 = 0.20

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
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<u>Compute P(cavity):</u>

P (cavity)

$= 0.108 + 0.012 + 0.072 + 0.008 = 0.20$

Inference by Enumeration (Self Study)

	toothache		*¬toothache*	
	catch	*¬catch*	*catch*	*¬catch*
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

<u>Compute P(cavity ∨ toothache):</u>

P (cavity ∨ toothache)

= 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
$\neg cavity$	0.016	0.064	0.144	0.576

<u>Compute conditional probability, P(¬*cavity* | *toothache*)</u>

$$P(\neg cavity \mid toothache) = \frac{P(\neg cavity \wedge toothache)}{P(toothache)}$$

$$= \frac{0.016+0.064}{0.108 + 0.012 + 0.016 + 0.064}$$

$$= 0.4$$

Inference by Enumeration (Self Study)

	$toothache$		$\neg toothache$	
	$catch$	$\neg catch$	$catch$	$\neg catch$
$cavity$	0.108	0.012	0.072	0.008
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<u>Compute conditional probability, P(*cavity* | *toothache*)</u>

$$P(cavity \mid toothache) = \frac{P(cavity \wedge toothache)}{P(toothache)}$$

$$= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064}$$

$$= 0.6$$

Complexity of Enumeration (Self Study)

- Worst case time: $O(d^n)$
  - Where d = max arity
  - And n = number of random variables
- Space complexity also $O(d^n)$
  - Size of joint distribution
- Prohibitive!

Independence

A and *B* are *independent* iff:

$$P(A \mid B) = P(A)$$

$$P(B \mid A) = P(B)$$

These two constraints are logically equivalent

Therefore, if *A* and *B* are independent:

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)} = P(A)$$

$$P(A \wedge B) = P(A)P(B)$$

Bayes Rule

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

A, B = events

$P(A|B)$ = probability of A given B is true

$P(B|A)$ = probability of B given A is true

$P(A), P(B)$ = the independent probabilities of A and B

Bayesian Network

- A simple, graphical notation for conditional independence assertions and hence for compact specification of full joint distributions.
- Syntax:
  - a set of nodes, one per variable
  - a directed, acyclic graph (link ≈ "directly influences")
  - a conditional distribution for each node given its parents: **P** (Xi | Parents (Xi))
- In the simplest case, conditional distribution represented as a conditional probability table (CPT) giving the distribution over X_i for each combination of parent values

Example Bayesian Net #1

A simple world with four random variables

- Weather, Toothache, Cavity, Catch
- Weather is independent of
  - other variables
- Toothache and Catch are conditionally independent given Cavity
- P(Toothache, Catch | Cavity) = P(Toothache | Cavity) . P(Catch | Cavity)
- Cavity is a direct cause of Toothache and Catch
- No direct relation between Toothache and Catch exists

ML here is marginalization. Not Machine learning!

Example Bayesian Net #1

A simple world with four random variables

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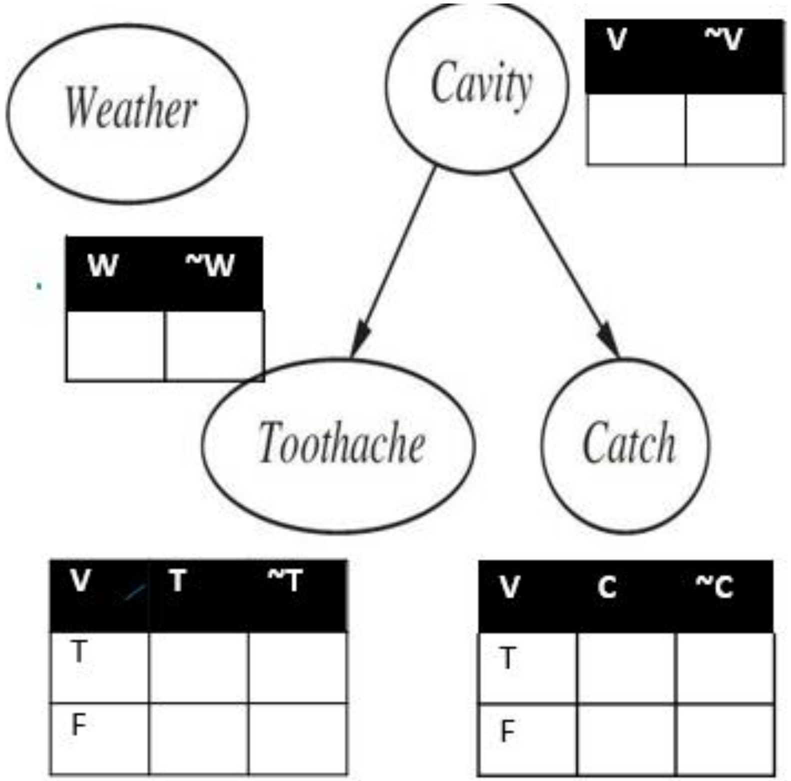
Example Bayesian Net #1

Question:

What is the probability of good weather, not having toothache, not having cavity and having catch?

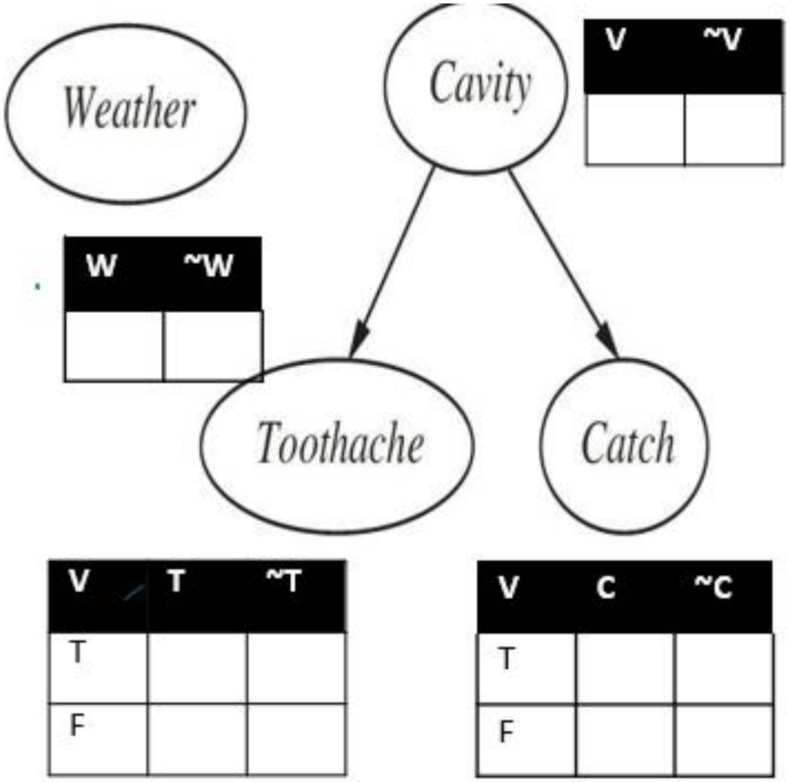
P(W,~T, ~V, C) = ?



Example Bayesian Net #1

P(W,~T, ~V, C) = ?

Answer:



Example Bayesian Net #2

A Burglary Alarm System:

- Fairly reliable on detecting a burglary
- Also responds to earthquakes
- Two neighbors John and Mary are asked to call you at work when Burglary happens and they hear the Alarm
- John nearly always calls when he hears the alarm, however sometimes confuses the telephone ring with alarm and calls then too
- Mary like loud music and often misses the alarm altogether
- **Problem**: Given the information that who has / has not called we need to estimate the probability of a burglary

Example Bayesian Net #2

A Burglary Alarm System:

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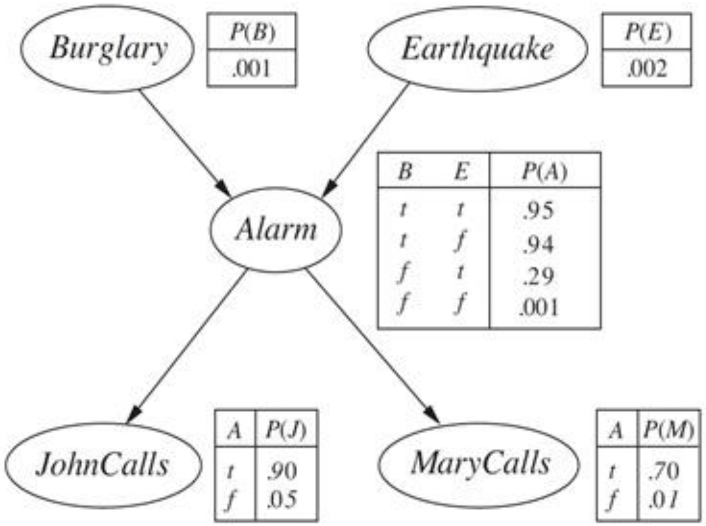
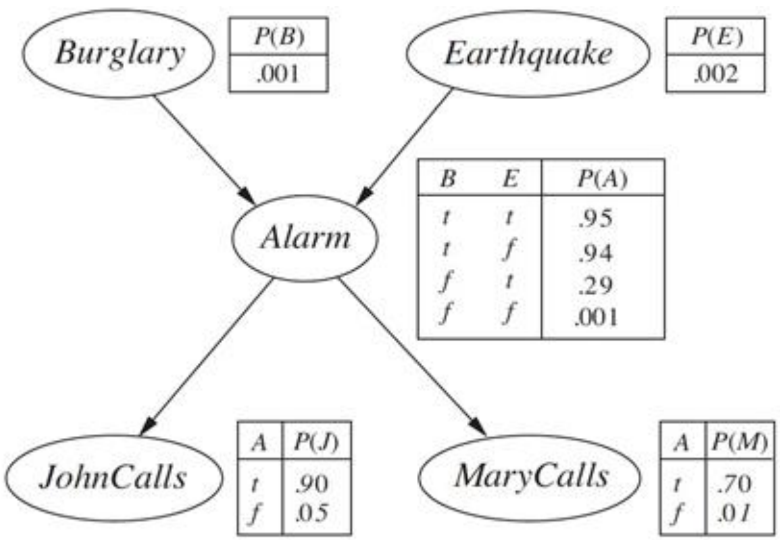


Examples

Calculate the probability that alarm has sounded, but neither burglary nor earthquake happened, and both John and Mary called.



$$P(j, m, a, \neg b, \neg e) = P(j \mid a)P(m \mid a)P(a \mid \neg b \wedge \neg e)P(\neg b)P(\neg e)$$

$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 = 0.000628$$

Example Bayesian Net #3

Traffic Prediction -Travel Estimation

- AI system reminds traveler regarding start time
- Travel plan is to reach BLR and the weather of BLR may influence the accommodation plans
- Traveler always take car to reach airport
- Car may be rerouted either due to road block or weekend traffic during working hours which delays the arrival to airport
- Bars are always observed to be full on weekends
- Authorities block roads to safe the processions
- Processions observed during festive season or due to the political rally.
- **Problem**: Given the information that there is a political rally expected estimate the probability of late arrival

Example Bayesian Net #3

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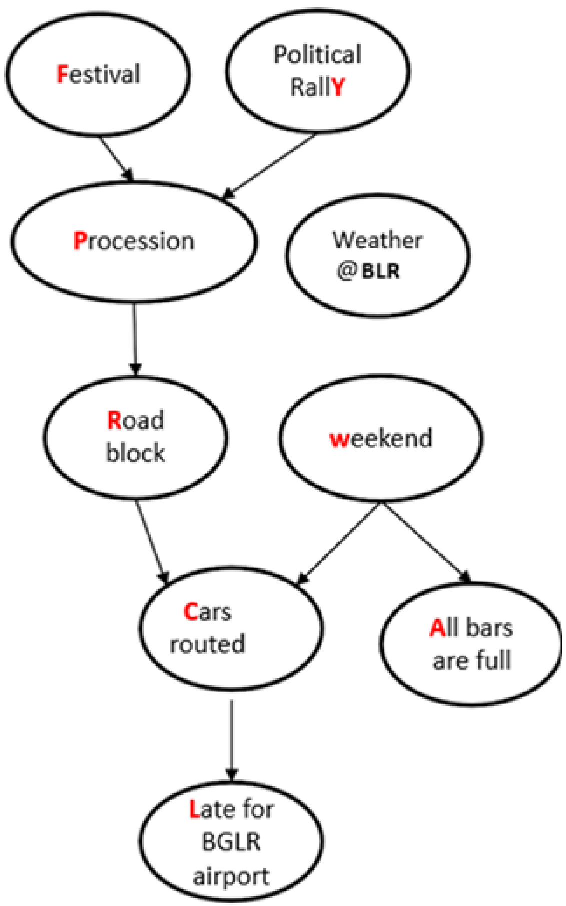
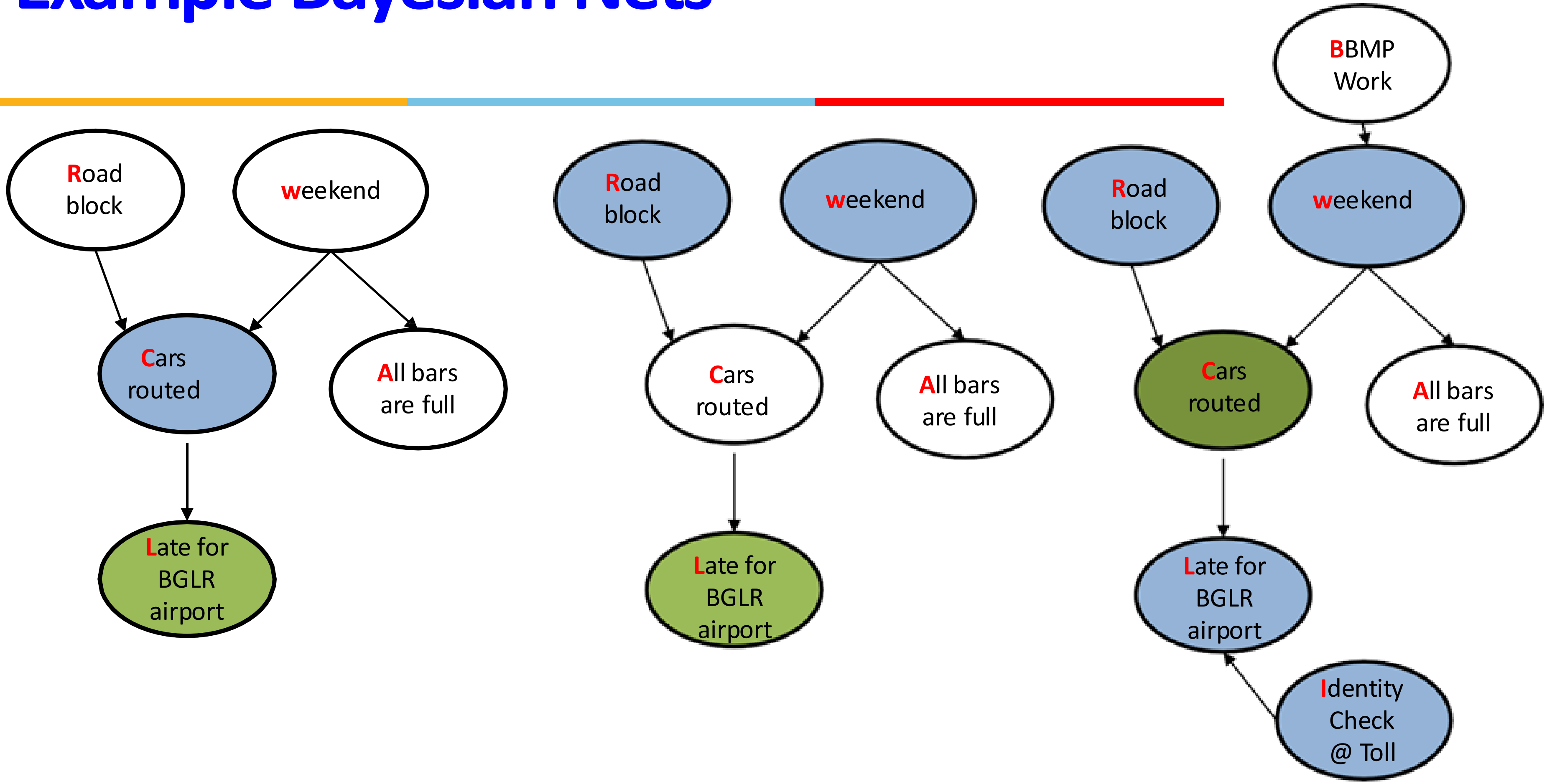
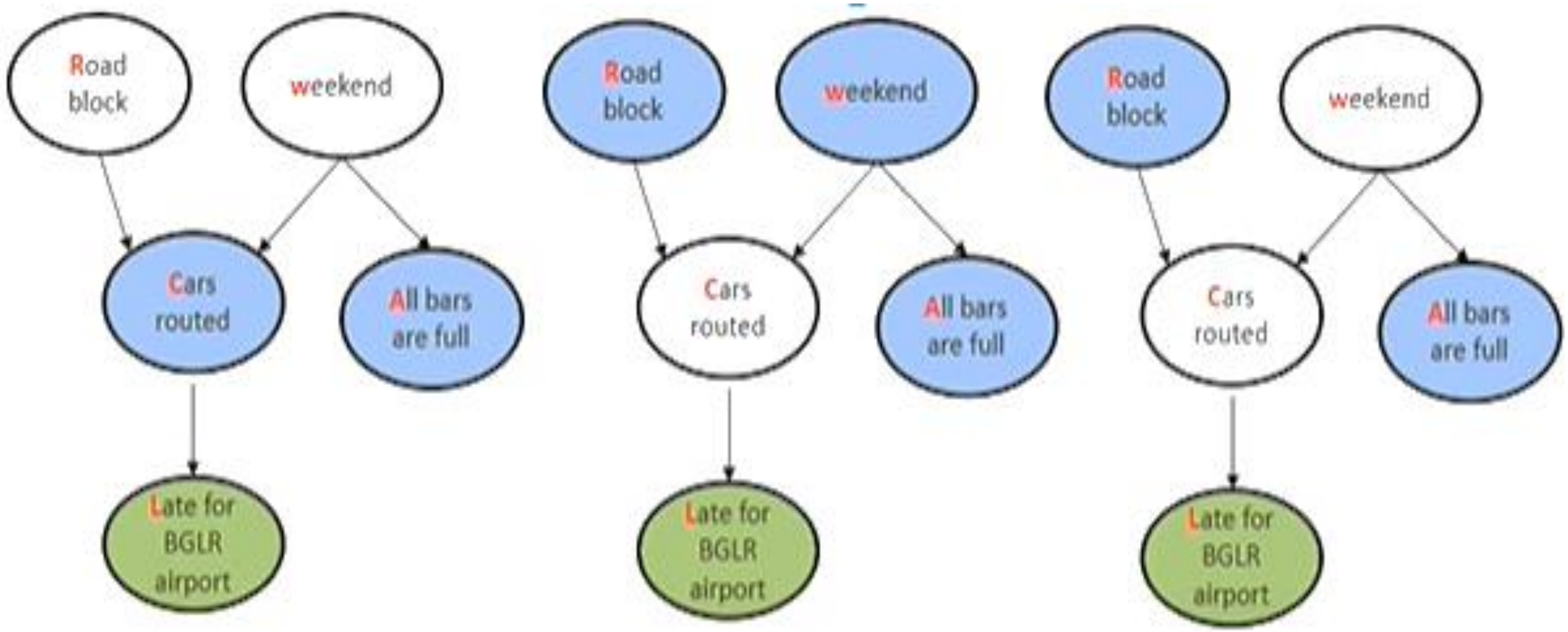


Example Bayesian Nets



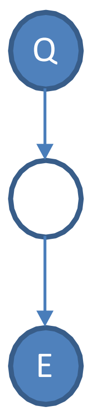
- A node is conditionally independent of its non-descendants given its parents
- A node is conditionally independent of all other nodes in the net , given its parents, children and
  - children's parents.

Example Bayesian Nets

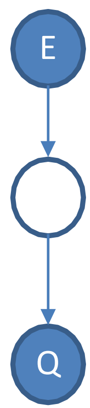


Belief Nets

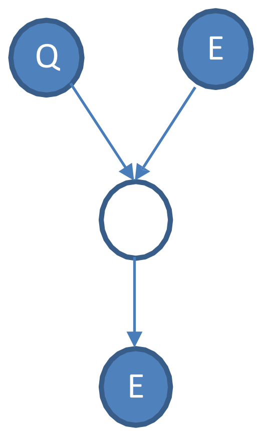
Diagnostic



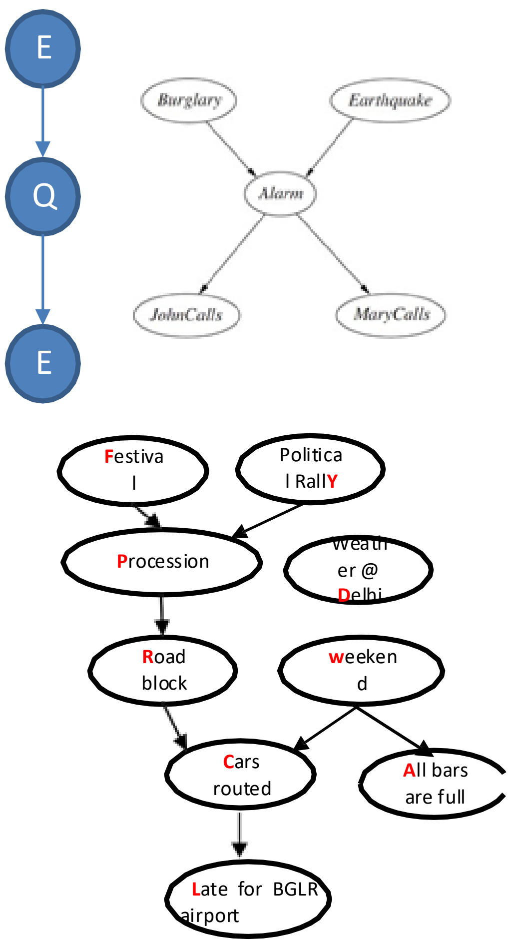
Causal



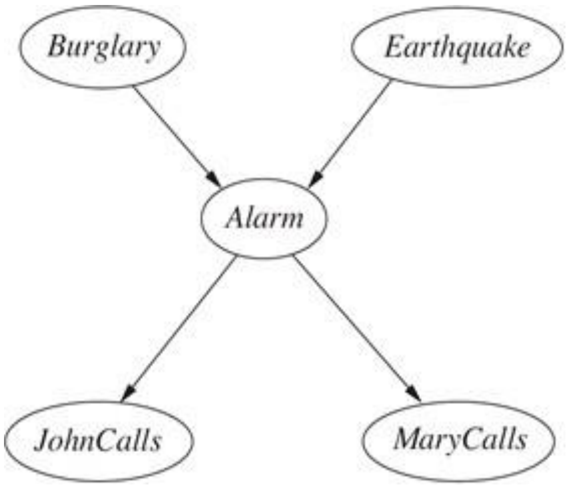
Inter-Casual



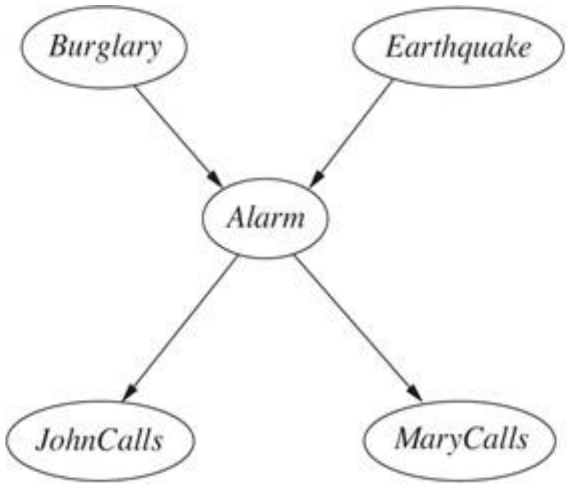
Mixed Inferences



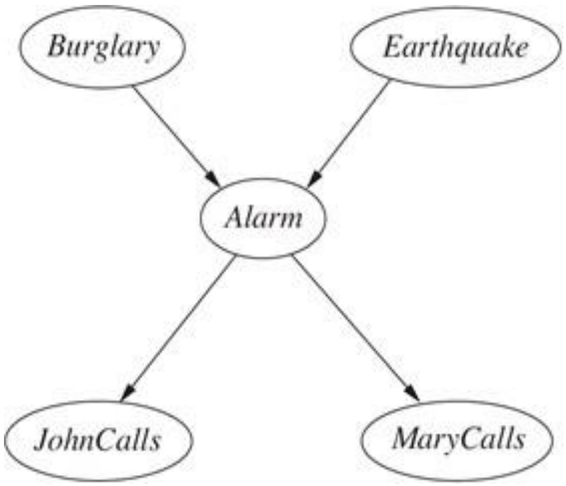
Belief Nets
(Identify Inference Type)



(1) P(J / B~A)



(2) P(E/BM)



(3) P(A/J)

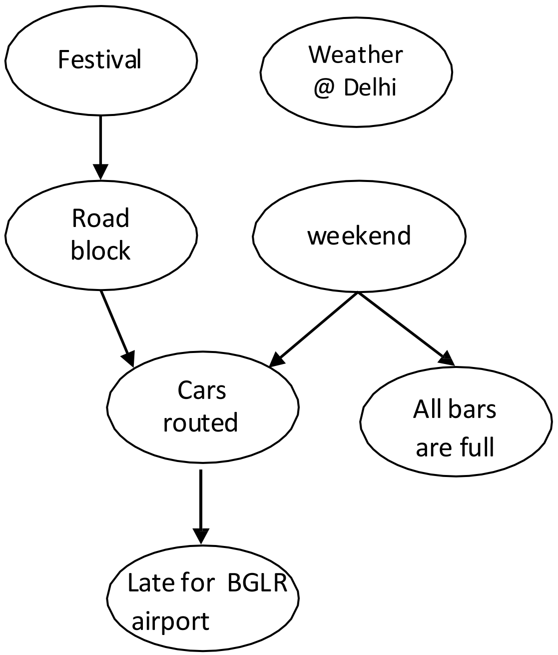
Inference with Bayesian Networks

- Exact Inference
  - By Enumeration
  - Variable Elimination
- Approximate Inference

Inferences in Bayesian Nets

Enumeration

Examples



1. Calculate the probability that arrival at airport was delayed during a weekend but there was no road block or festival and car was not routed anywhere.

   P(L,W,~R,~F,~C) ? Joint

2. What is the probability that it is a festival season given cars where routed?

   P(F|C) ? Conditional

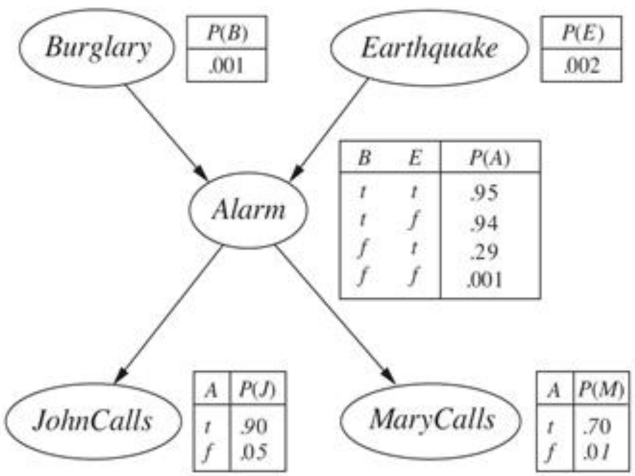
3. What is the probability that car arrived late at airport given it's a festival day?

   P(L| F) ? Conditional

Examples

What is the probability that Burglary happened given John & Mary called the police?

$$P(B \mid JM) + P(\sim B \mid JM) = 1$$



$$\frac{P(BJM)}{P(JM)} + \frac{P(\sim BJM)}{P(JM)} = 1$$

$$\frac{1}{P(JM)}\,[P(BJM) + P(\sim BJM)] = 1$$

Let $\alpha = \dfrac{1}{P(JM)}$

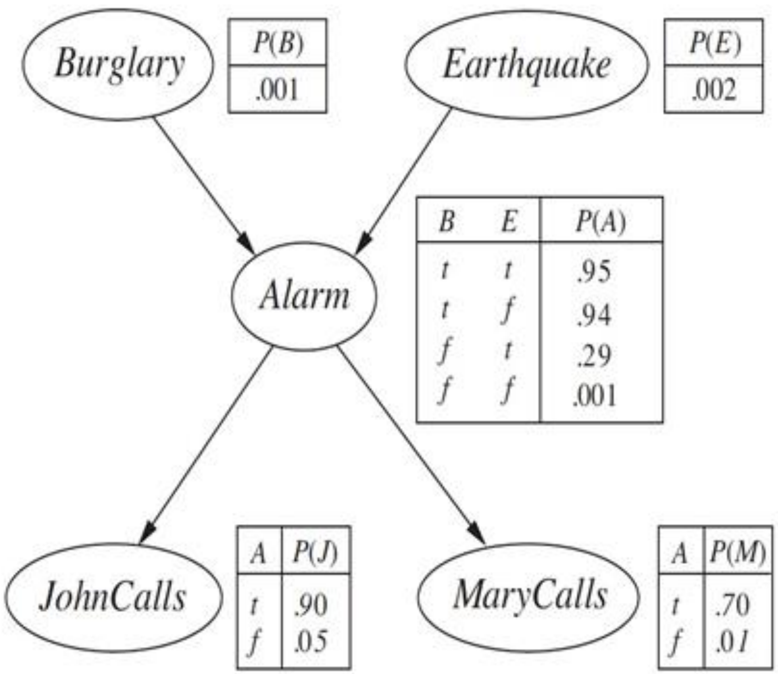
$$P(A) + P(\sim A) = 1$$

$P(A \mid B) = \dfrac{P(AB)}{P(B)}$ Cond Prob

Then, $\alpha = \dfrac{1}{P(BJM)+P(\sim BJM)}$ --- (1)

Examples

What is the probability that Burglary happened given John & Mary called the police?



$$P(B \mid JM) = \frac{P(BJM)}{P(JM)} \quad \text{--- (2)}$$

$Put\ (1)\ in\ (2)$:

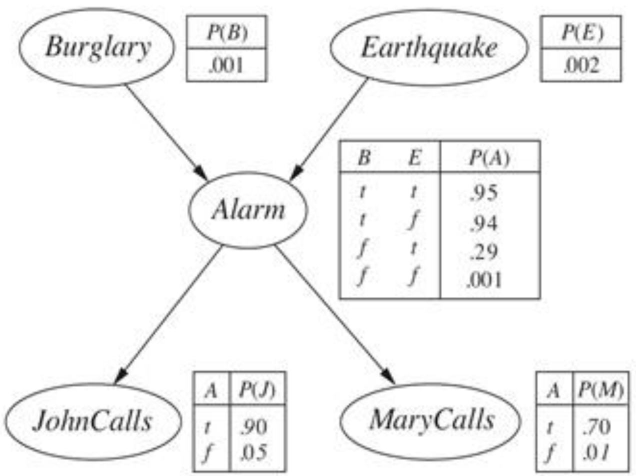
$$P(B \mid JM) = \frac{P(BJM)}{P(BJM) + P(\sim BJM)}$$

☺ Now query is in joint probability!

Calculate P(BJM) and B(~BJM) and substitute the values in above to get the answer.

Examples

What is the probability that Burglary happened given John & Mary called the police?



$\mathrm{P(BJM)} = ?$

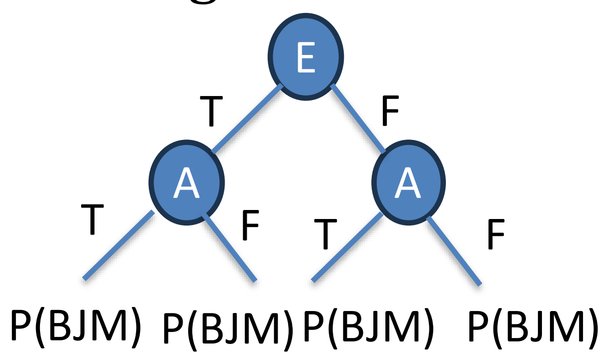
Here, A and E are hidden variables!

$\mathrm{P(BJM)} = \sum_A \sum_E \mathrm{JMABE}$

Apply Chain rule:

$$\mathrm{P(BJM)} = \sum_A \sum_E P(J\,|A) * P(M\,|A) * P(A\,|\,BE) * P(B) * P(E)$$

Marginalization!



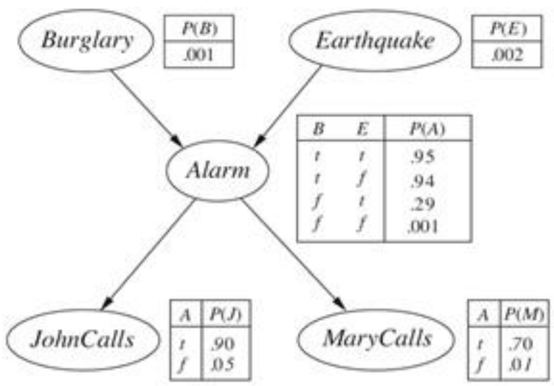
Remove $\sum_E$ *by assigning* T, F:

$$\mathrm{P(BJM)} = \sum_A \begin{array}{l} [P(J\,|A) * P(M\,|A) * P(A\,|\,BE) * P(B) * P(E)] + \\ [P(J\,|A) * P(M\,|A) * P(A\,|\,B{\sim}E) * P(B) * P({\sim}E)] \end{array}$$

Calculate the above 4 values and add.

Examples

What is the probability that Burglary happened given John & Mary called the police?



Remove $\sum_A$ *by assigning* T, F:

$$
\begin{gathered}
\mathrm{P(BJM)} = \\
[P(J\,|A) * P(M\,|A) * P(A \,|\, BE) * P(B) * P(E)] + \\
[P(J\,|{\sim}A) * P(M\,|{\sim}A) * P({\sim}A \,|\, BE) * P(B) * P(E)] + \\
[P(J\,|A) * P(M\,|A) * P(A\,|B{\sim}E) * P(B) * P({\sim}E)] + \\
[P(J|{\sim}A) * P(M\,|{\sim}A) * P({\sim}A \,|\, B\,{\sim}E) * P(B) * P({\sim}E)]
\end{gathered}
$$

$$
\mathrm{P(BJM)} = \begin{gathered}
[0.90 * 0.70 * 0.95 * 0.001 * 0.002] + \\
[0.05 * 0.01 * [1 - 0.95] * 0.001 * 0.002] + \\
[0.90 * 0.70 * 0.94 * 0.001 * [1 - 0.002]] + \\
[0.05 * 0.01 * [1 - 0.94] * 0.01 * [1 - 0.002]]
\end{gathered}
$$

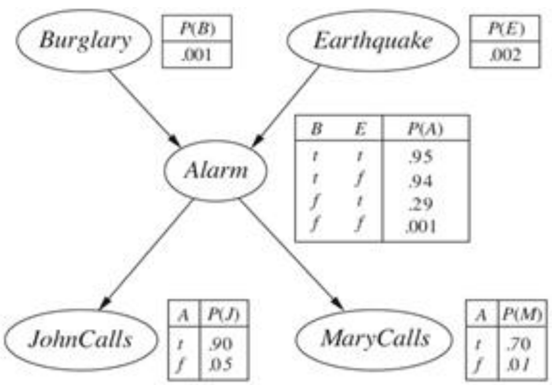
$\mathrm{P(BJM)} = 0.000001197 + 0.00000000005 + 0.000592 + 0.0000002994$

≈ 0.0005935

We still do not know about P(~BJM) !

Examples

What is the probability that Burglary happened given John & Mary called the police?



$$\mathrm{P(\sim BJM)} =$$
$$[P(J\,|A) * P(M\,|A) * P(A\,|\sim BE) * P(\sim B) * P(E)] +$$
$$[P(J\,|\sim A) * P(M\,|\sim A) * P(\sim A\,|\sim BE) * P(\sim B) * P(E)] +$$
$$[P(J\,|A) * P(M\,|A) * P(A\,|\sim B\sim E) * P(\sim B) * P(\sim E)] +$$
$$[P(J|\sim A) * P(M\,|\sim A) * P(\sim A\,|\sim B\sim E) * P(\sim B) * P(\sim E)]$$

$$\mathrm{P(\sim BJM)} =$$
$$[0.90 * 0.70 * 0.29 * [1-0.001] * 0.002] +$$
$$[0.05 * 0.01 * [1-0.29] * [1-0.001] * 0.002] +$$
$$[0.90 * 0.70 * 0.001 * [1-0.001] * [1-0.002]] +$$
$$[0.05 * 0.01 * [1-0.001] * [1-0.001] * [1-0.002]]$$

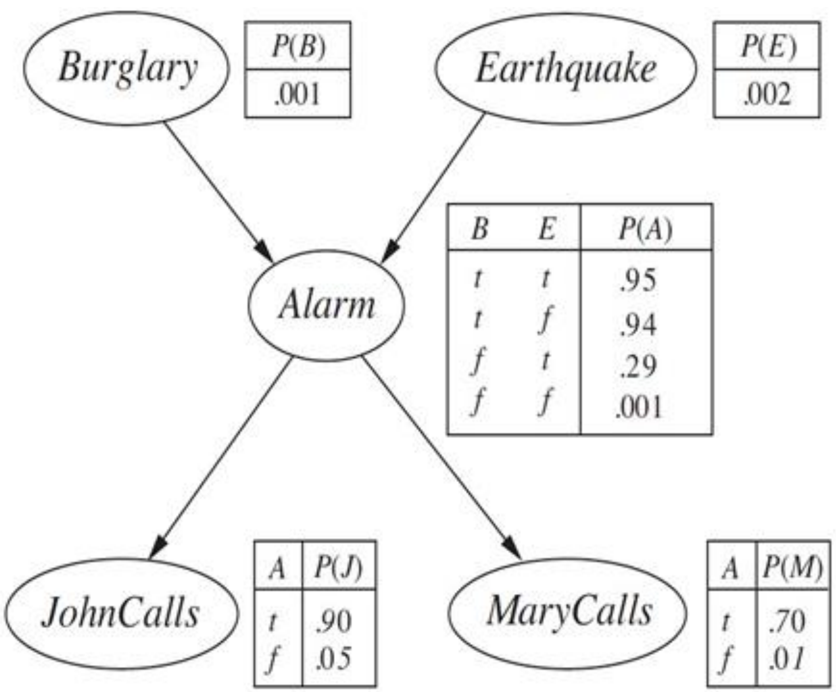
$$\mathrm{P(\sim BJM)} = 0.0003650346 + 0.00000070929 + 0.00062811126 + 0.000498002499$$

$$\approx 0.001492$$

$$P(B\,|\,JM) = \frac{P(BJM)}{P(BJM) + P(\sim BJM)} = P(B\,|\,JM) = \frac{0.0005935}{0.0005935+0.001492} = 0.2847$$

Exercise

- What is the probability that John calls given earthquake occurred?

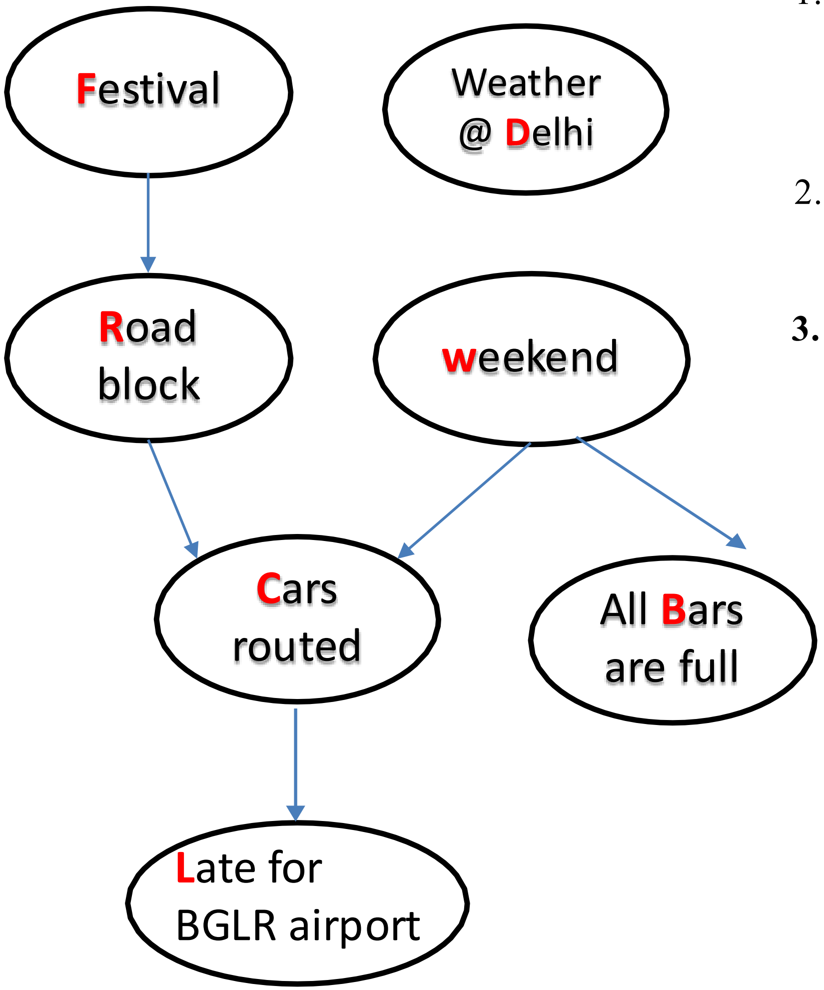


Inferences in Bayesian Nets

Variable Elimination

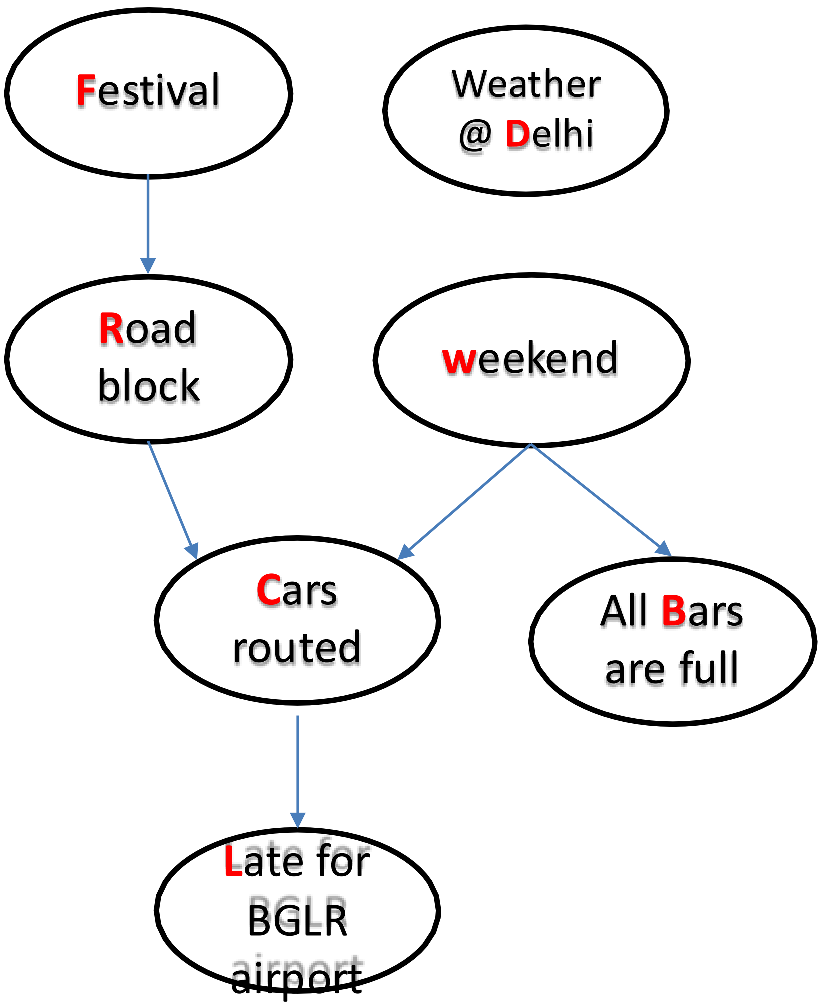
Reduce Guaranteed Independent nodes

D-Connectedness Vs D-Separation



1. Each variable is conditionally independent of its non-
descendants, given its parents
2. Eliminate the hidden variables that is neither a query nor an evidence
3. **Two variables are d-separated if they are conditionally independent given evidences**

Try it & Test

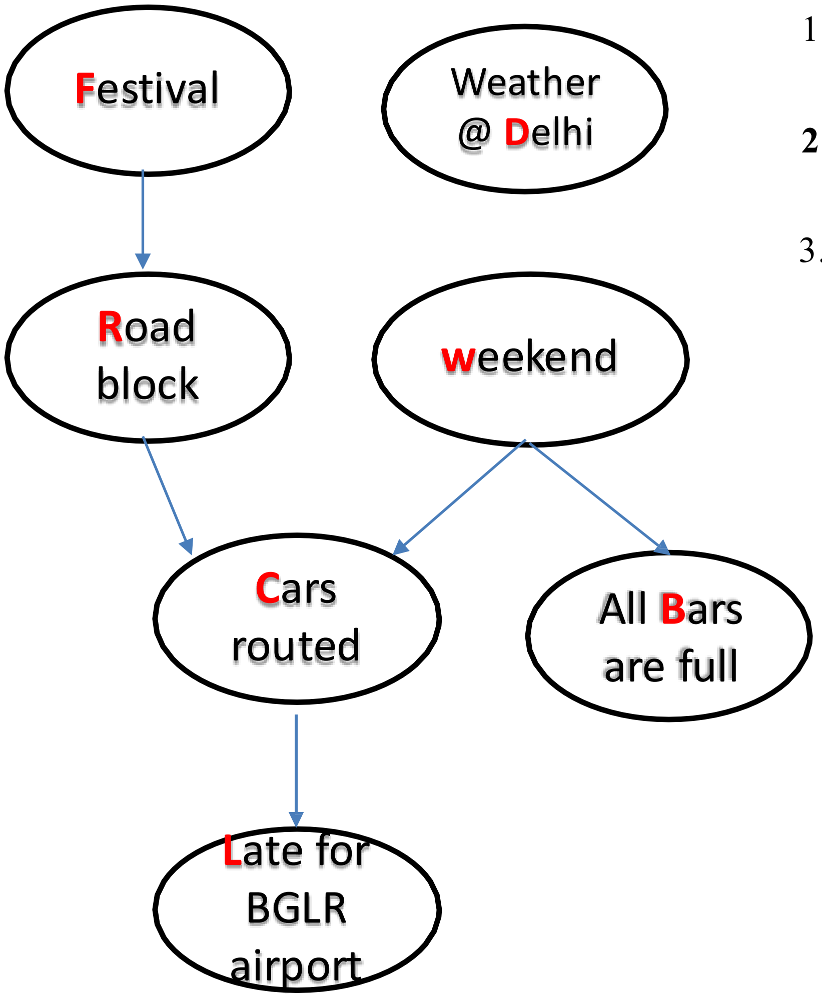


X	Y	Evidence Z	d-sep?
F	W	C	No
L	W	R	No
R	**L**	**C**	**Yes**
B	R	C	No

- P(R | L , C) = P(R| L)

R & L are d-separated i.e., conditionally independent given C

Variable Elimination



1. Each variable is conditionally independent of its non-descendants, given its parents
2. **Eliminate the hidden variables that is neither a query nor evidence**
3. Two variables are d-separated if they are conditionally independent given evidences

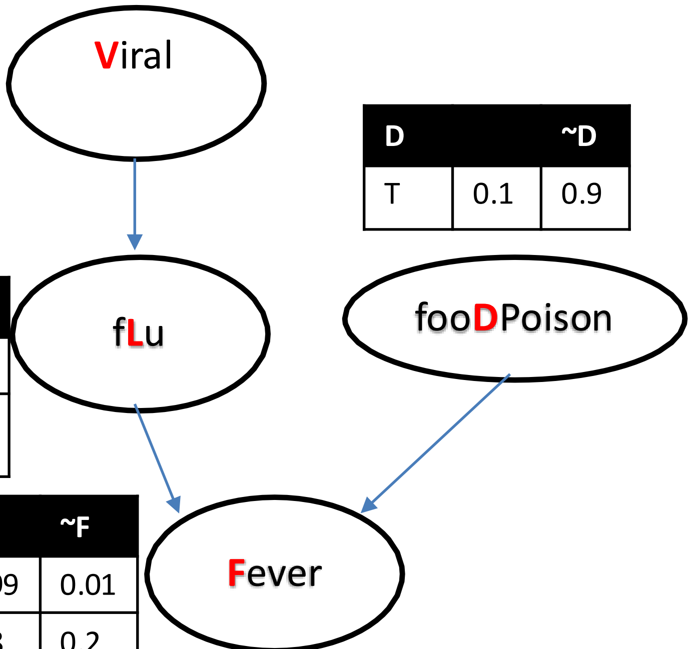
➢ $P(B) = \sum_{L, B, W, R, F} P(L, C, B, W, R, F)$
$= \sum_{L} \sum_{B} P(L|C) \,.\, P(B \mid W) \,.\, \sum_{W} P(C| W, R). \sum_{R} P(R \mid F) \,.\, \sum_{F} P(F)$
$= \mathbf{P(B \mid W)}$

All other variables are hidden w.r.t to B as (L, C, R, F) are neither evidence nor query nor (L, C, R, F) ∈ Ancestors(W, B)

This is variable elimination example targeting irrelevant nodes

D-Separation in Inference

V		~V
T	0.6	0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

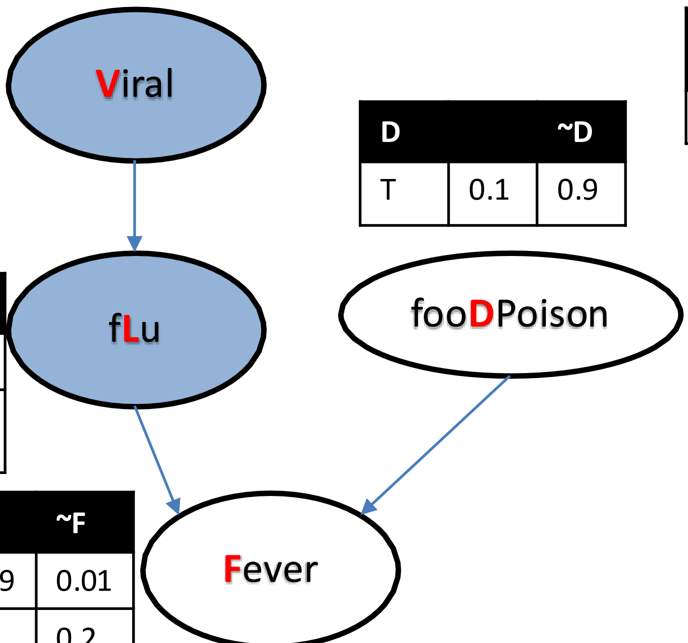
X	Y	Evidence Z	d-sep?
V	F	L	Yes
V	D	L	Yes

- P(V | F , L)
- P(V | D , L)

Inference

Variable Elimination: V

V		~V
T	0.6	0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V		~V
T	0.6	0.4

V	L	~L
T	0.8	0.2
F	0.01	0.99

V	L	
T	T	0.48
T	F	0.12
F	T	0.004
F	F	0.396

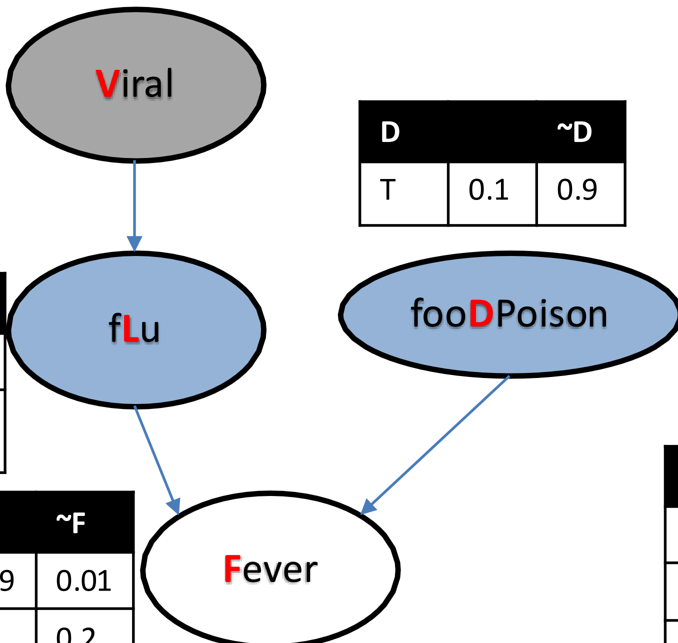
L	
T	0.484
F	0.516

P(V)
P(L|V)
P(D)
P(F|L,D)

Inference

Variable Elimination: L,D

V		~V
T	0.6	0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

P(L)
P(D)
P(F|L,D)

L	
T	0.484
F	0.516

D		~D
T	0.1	0.9

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

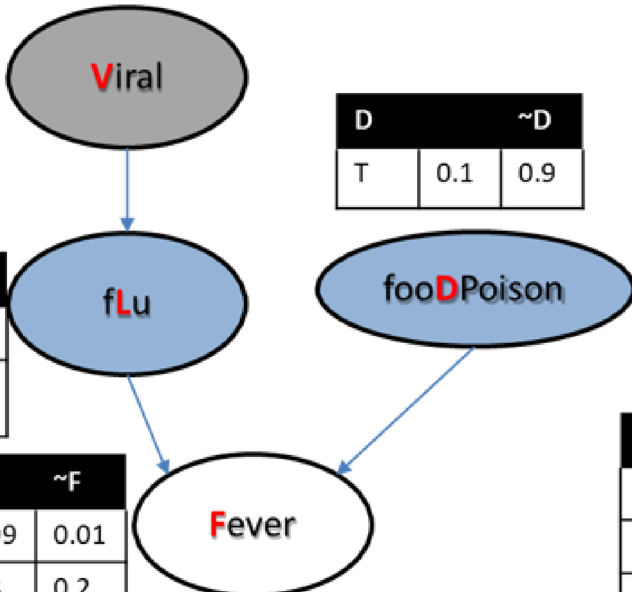
D	L	F	
T	T	T	
T	F	T	
F	T	T	
F	F	T	
T	T	F	
T	F	F	
F	T	F	
F	F	F	

L	F	

F	

Inference

V		~V
T	0.6	0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

P(L)
P(D)
P(F|L,D)

L	
T	0.484
F	0.516

D		~D
T	0.1	0.9

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

D	L	F	
T	T	T	0.048
T	F	T	0.04128
F	T	T	0.34852
F	F	T	0.02322
T	T	F	0.00048
T	F	F	0.01032
F	T	F	0.087
F	F	F	0.44118

L	F	
T	T	0.39652
F	T	0.0645
T	F	0.08748
F	F	0.4515

F	
T	0.46102
F	0.53898

Exercise

Consider the below Bayesian Network and answer the following questions:

Most of the WILP students are fans (F) of cricket irrespective of their gender. With the new season of IPL (Indian Premier League) having started on the exam month almost every cricket fans spend time to watch(W) the live play. Sometimes being a parent (P) reduces the probability of watching the IPL live season. A likely consequence of watching matches is reduced concentration(C) on the following day/s. A consequence of the reduced concentration is increased stress(S) with work environment leading to reduced productivity (D) in project. Lack of concentration might also be caused by viral (V) infection, which is common in this rainy season(R). WILP students have the comprehensive exams and reduced concentration would reduce the probability of good grades (G) in the exam which reflects the performance of students in examination. Assume an AI agent is fed this information and it answers to certain queries that can be inferred. Assume all the events(conditional or unconditional) are equally likely to occur:

1) Construct Bayesian network.
2) What is the chance that "an ardent fan of cricket who is a parent of two kids, never misses an IPL match, doesn't get stressed in work environment, is affected by viral infection and performs well in the comprehensive examination"?
3) Performance of in the examination is independent of stress in work environment given its known that the student is affected by viral infection. Check for D-separation.

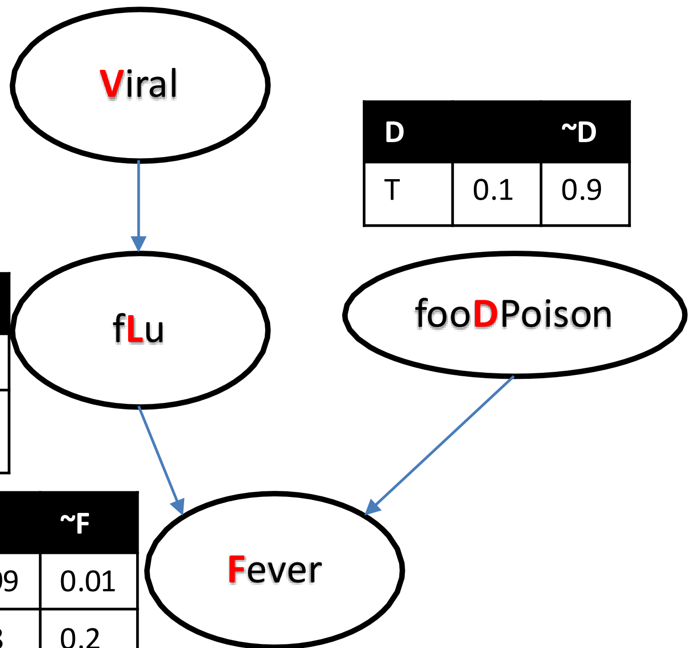
Approximate Inferences in Bayesian Nets Introduction

Prior Sampling

Sample Generation by Randomization

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F

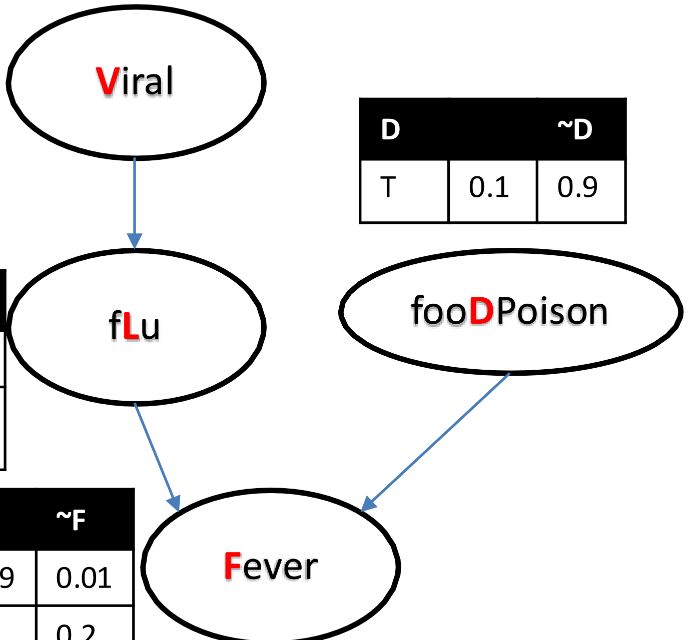
0.3, 0.2, 0.6, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.9,0.55…………..

Prior Sampling

Sample Generation by Randomization

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

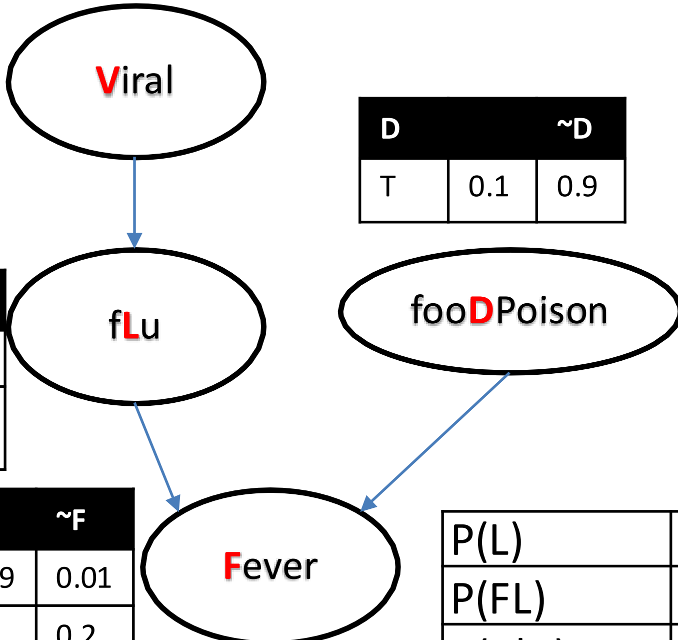
V	L	D	F
T	T	F	T
F	F	F	F
T	F	F	T
F	T	F	T
..			
…..			

0.3, 0.2, 0.6, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.9,0.55…………..

Prior Sampling

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F
T	T	F	T
F	F	F	F
T	F	F	T
F	T	F	T
T	T	F	T
T	F	F	F
F	F	F	T
T	F	F	F

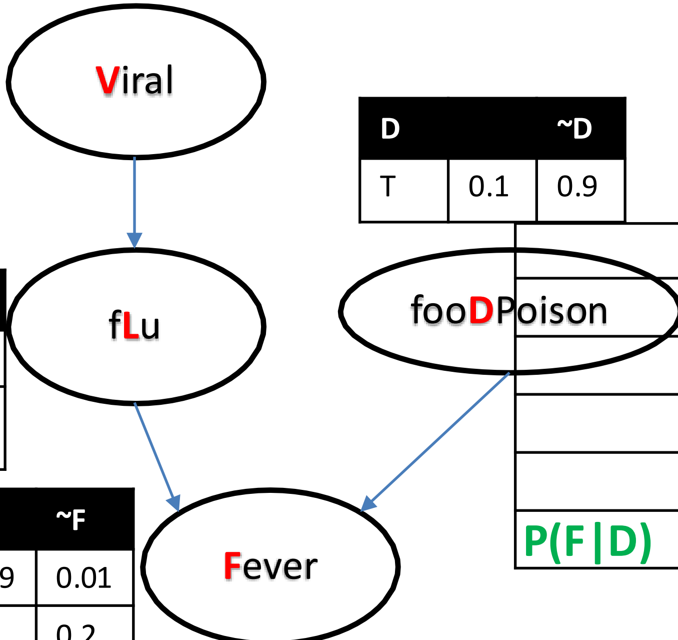
P(L)	= 3/8
P(FL)	= 3/8
P(L|F)	= 3/5
P(~V|F)	= 2/5
P(L|V~F)	= 0
P(F|D)	= ?????

Rejection Sampling

Sample Generation by Randomization

V		~V
T	0.6	

0.4



V	L	~L
T	0.8	0.2
F	0.01	0.99

D		~D
T	0.1	0.9

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

P(F|D)	= ?????

V	L	D	F
…..			

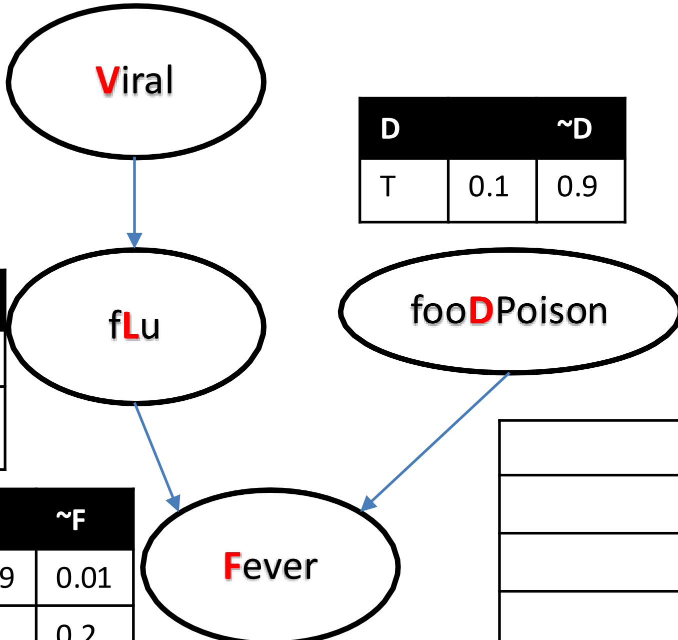
0.3, 0.2, 0.6, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.9,0.555,0.38…………..

Rejection Sampling

Sample Generation by Randomization

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F
T	T	F	
T	T	F	
T	T	F	
F	F	T	F
T	F	T	T
…..			

P(F|D)	= ?????

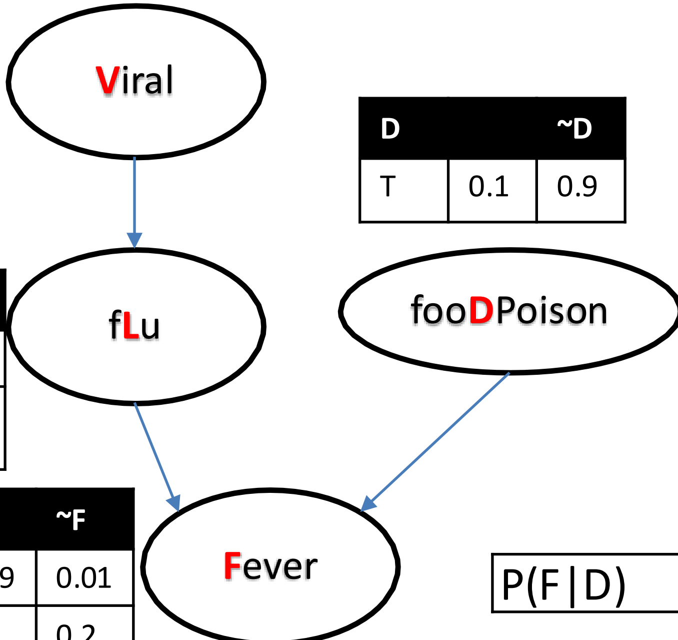
0.3, 0.2, 0.6, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.9,0.555,0.38…………..

Rejection Sampling

Inference

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

P(F|D)	= 5/8

V	L	D	F
T	T	T	T
F	F	T	F
T	F	T	T
F	T	T	T
T	T	T	T
T	F	T	F
F	F	T	T
T	F	T	F

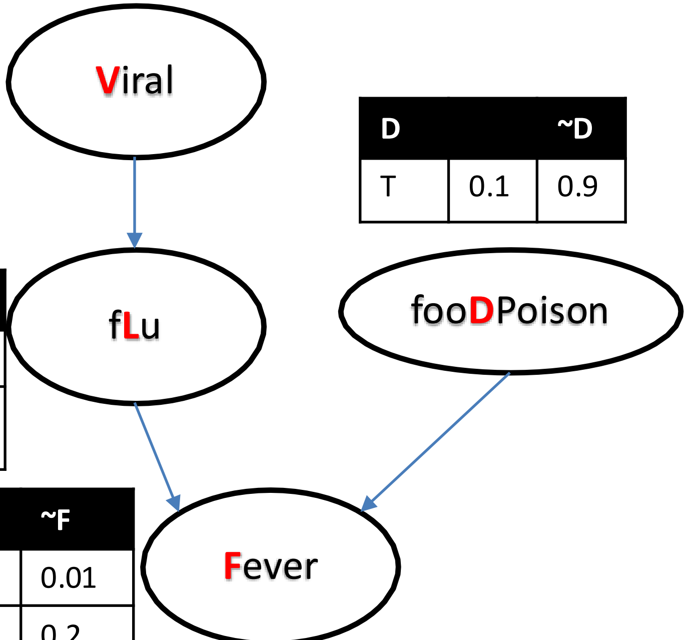
Likelihood Weighing

Sample Generation by Randomization

V		~V
T	0.6	

0.4

D		~D
T	0.1	0.9



V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F	wgt
F		T		
F		T		
F		T		
F		T		
F		T		
F		T		
F		T		

P(~F|D,~V) = 0.04 / 7*0.04

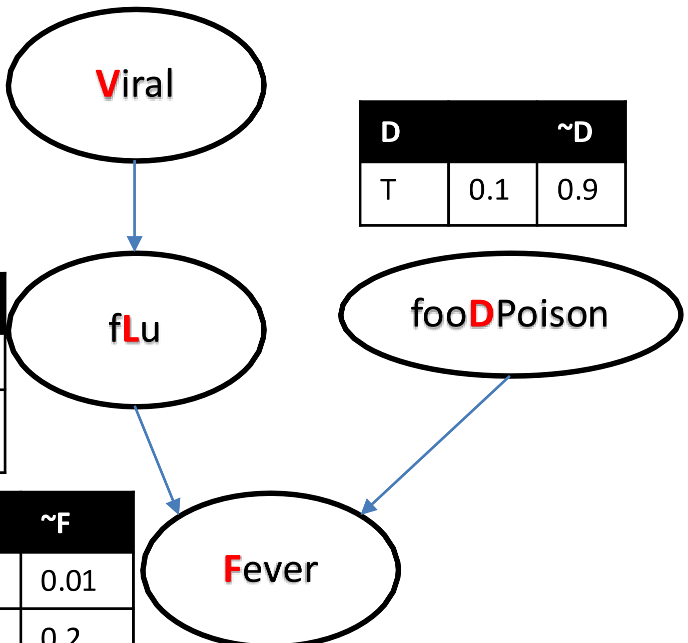
0.3, 0.2, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.99,,……….

Likelihood Weighing

Sample Generation by Randomization

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F	wgt
F	F	T	T	0.4*1* 0.1 *1=
F	F	T	T	
F	F	T	T	
F	F	T	T	
F	F	T	T	
F	T	T	T	
F	T	T	F	

P(~F|D,~V) = 0.04 / 7*0.04

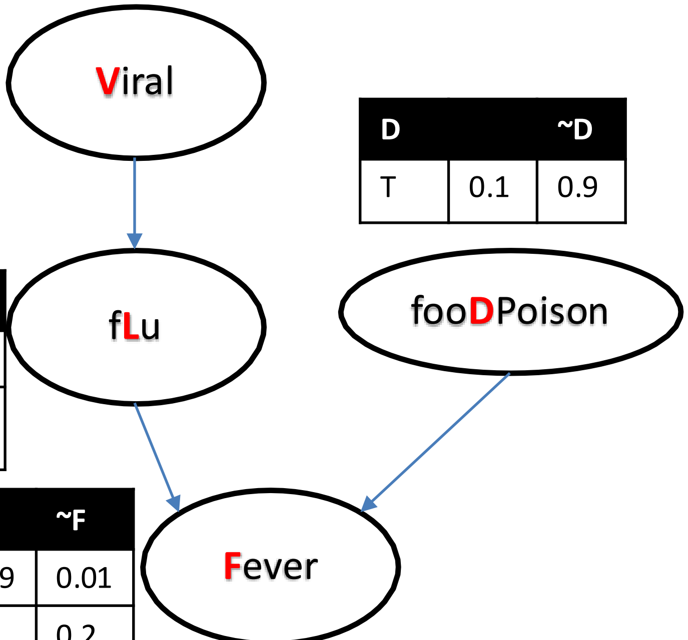
0.3, 0.2, 0.58, 0.73, 0.87, 0.15, 0.6, 0.57, 0.85, 0.12, 0.004, 0.93 , 0.0002, 0.99,,……….

Likelihood Weighing

Inference

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F	wgt
F	F	T	F	0.4*1* 0.1 *1=
F	T	T	T	0.4*1* 0.1 *1=
F	F	T	T	0.4*1* 0.1 *1=
F	F	T	F	0.4*1* 0.1 *1=

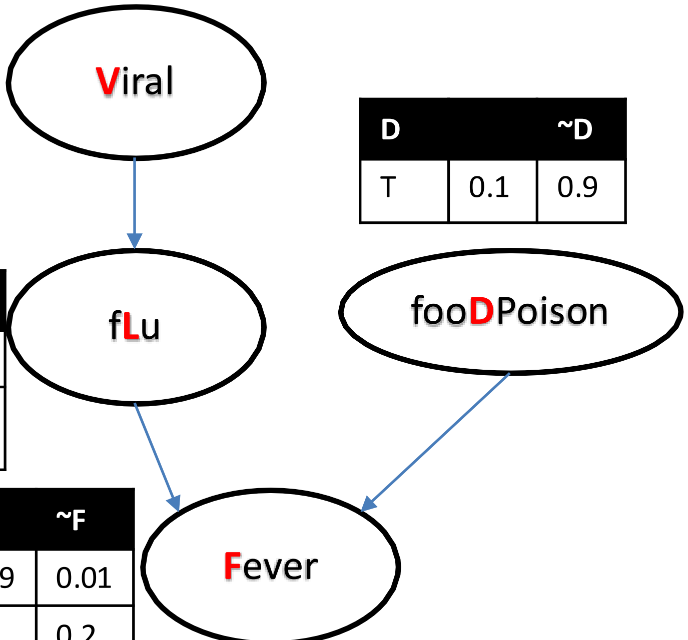
$$P(F|D,\sim V) = 0.04+0.04 \ / \ 4*0.04$$

Likelihood Weighing

Inference (Another Example)

V		~V
T	0.6	

0.4



D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

V	L	D	F	wgt
F	F	T	F	
F	F	T	T	
F	F	T	T	
T	F	T	F	

$$P(F|D,\sim L) \;= 0.099+0.099 \;\; / \; (3*0.099 + 0.02)$$

Likelihood Weighing

Inference

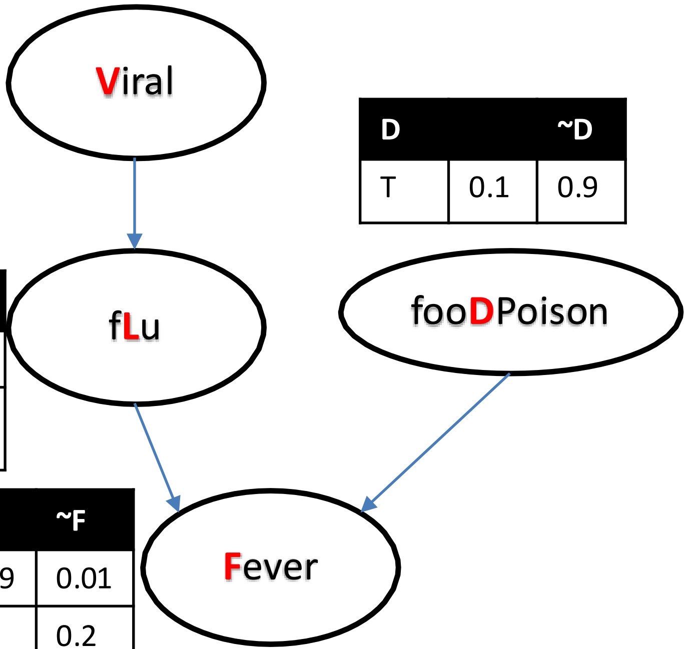
V		~V
T	0.6	

0.4

D		~D
T	0.1	0.9

V	L	~L
T	0.8	0.2
F	0.01	0.99

L	D	F	~F
T	T	0.99	0.01
T	F	0.8	0.2
F	T	0.8	0.2
F	F	0.05	0.95

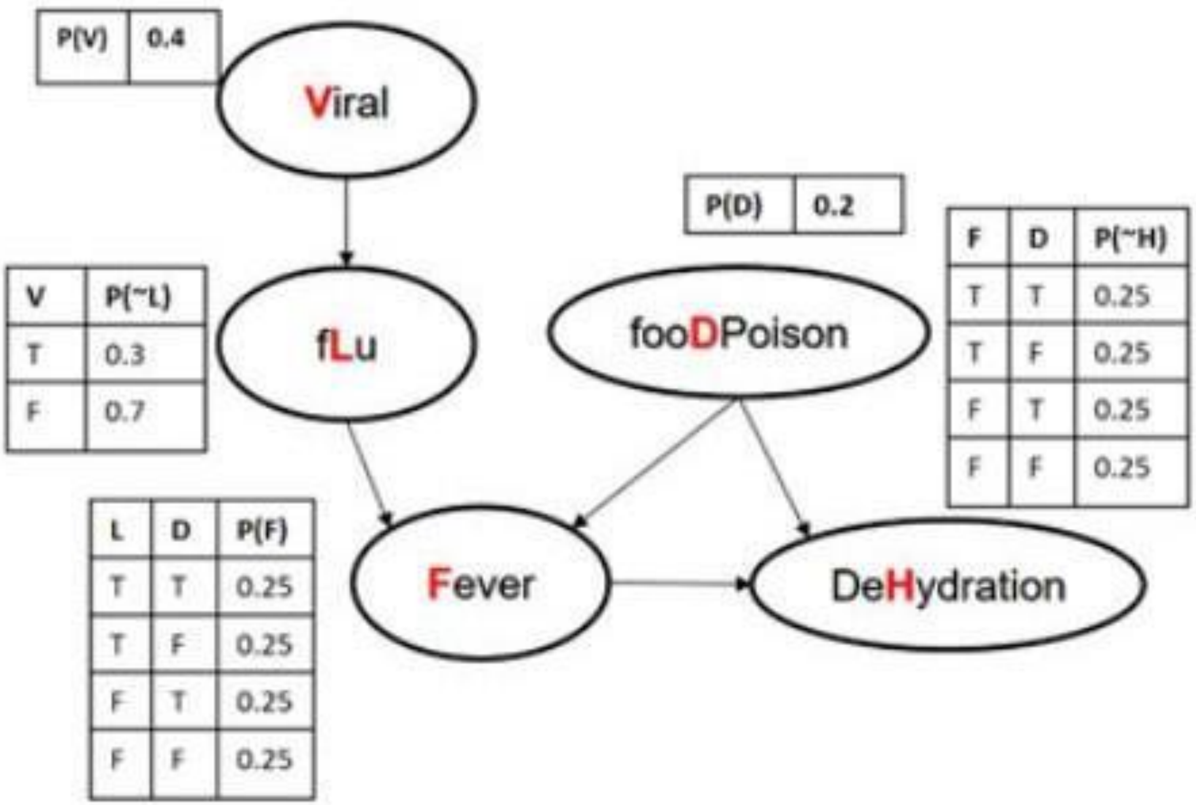


V	L	D	F	wgt
F	F	T	F	1*0.99* 0.1 *1=
F	F	T	T	1*0.99* 0.1 *1=
F	F	T	T	1*0.99* 0.1 *1=
T	F	T	F	1*0.2* 0.1 *1=

$$P(F|D,\sim L) = 0.099+0.099 \;/\; (3*0.099 + 0.02)$$

Exercise Problems

Consider the below Bayesian Network and answer the following questions:



Exact Inference: **What is the chance that a person doesn't get fever given the evidence that his/her blood test results show viral infection and severe dehydration?**

Approximate inference– Prior Sampling, Rejection sampling, likelihood weighing,

"0.3,0.6,0.2,0.1,0.7,0.5,0.5,0.25,0.45,0.85,0.35,0.9,0.15,0.65,0.51,0.2,0.7,0.10,0.6, 0.8"

Exercise Problem 2

Consider the below Bayesian Network and answer the following questions:

Most of the WILP students are fans (F) of cricket irrespective of their gender. With the new season of IPL (Indian Premier League) having started on the exam month almost every cricket fans spend time to watch(W) the live play. Sometimes being a parent (P) reduces the probability of watching the IPL live season. A likely consequence of watching matches is reduced concentration(C) on the following day/s. A consequence of the reduced concentration is increased stress(S) with work environment leading to reduced productivity (D) in project. Lack of concentration might also be caused by viral (V) infection, which is common in this rainy season(R). WILP students have the comprehensive exams and reduced concentration would reduce the probability of good grades (G) in the exam which reflects the performance of students in examination. Assume an AI agent is fed this information and it answers to certain queries that can be inferred. Assume all the events(conditional or unconditional) are equally likely to occur:

1) Construct Bayesian network.
2) What is the chance that "an ardent fan of cricket who is a parent of two kids, never misses an IPL match, doesn't get stressed in work environment, is affected by viral infection and performs well in the comprehensive examination"?
3) Performance of in the examination is independent of stress in work environment given its known that the student is affected by viral infection. Check for D-separation.

Citations and References

- Few content for these slides may have been obtained from prescribed books and various other source on the Internet
- I hereby acknowledge all the contributors for their material and inputs and gratefully acknowledge people others who made their course materials freely available online.
- .I have provided source information wherever necessary
- This is not a full fledged reading materials. Students are requested to refer to the textbook w.r.t detailed content of the presentation deck that is expected to be shared over e-learning portal - taxilla.
- I have added and modified the content to suit the requirements of the class dynamics & live session's lecture delivery flow for presentation

- **Slide Source / Preparation / Review:**
- From BITS Pilani WILP**:** Prof.Raja vadhana, Prof. Indumathi, Prof.Sangeetha
- From BITS Oncampus & External : Mr.Santosh GSK

Thank You for your time & attention !

Contact : parthasarathypd@wilp.bits-pilani.ac.in